Phizura: On-the-Fly Music Recording with Method Proxies

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Abstract
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1. Introduction

This paper introduces the Phizura Playground, a tool that extends the Pharo Playground with the capability to record the code written and evaluated within it. The recording system is implemented using Method Proxies [1].

The Phizura Playground has been designed to work with Coypu[2], a library and dialect for on-the-fly music programming within the Pharo environment. Coypu enables live coding performances in which musical structures are generated and modified interactively.

Capturing a musical performance without recording the audio output into large files has been a challenge since the early versions of Coypu. A first recording mechanism was implemented by Redwane Engels during an internship, and was demonstrated at ESUG 2024 in Lille.

This initial solution relied on the Pharo Announcer and Announcement classes, which implement the observer design pattern. Despite working reliably, this approach required intrusive modifications to the Coypu codebase and made it harder to extend the system with new behaviors. In recent months, we have reimplemented the recorder using Method Proxies. This redesign removes instrumentation from the system and provides a more flexible framework for extending live coding performances.

2. Objectives

While developing Coypu, we observed that the Pharo language and environment provide the means to easily record the code written and evaluated during a musical performance. This feature is novel in the live coding community and allows long performances to be stored in very small text files. These recordings can then be replayed by other musicians using the same audio server [3] and sound synthesis setup, or with a different *orchestra*[1]. Such lightweight text files enable easy sharing between users and allow concerts to be reproduced even in the absence of the human performer.

We initially implemented a recording mechanism in Coypu based on Announcers and Announcements. However, this solution was intrusive and tightly coupled to the Coypu codebase, making it difficult to extend and maintain, although it functioned as expected.

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[1]We use the term orchestra denotes the set of sound synthesis definitions, instruments, and audio effects used to render a score, following Csound terminology.

To address these limitations, we reimplemented the recorder using Method Proxies. This approach cleanly separates instrumentation from performance-related methods, resulting in a more modular and extensible design.

3. Design and Implementation

We extended the Pharo Playground to capture live evaluations. The extension adds a hook before expression execution. When recording is active, the hook sends the expression and a timestamp to a handler. The handler stores each evaluation with its `DateAndTime now`.

During replay, the system computes the duration between consecutive timestamps and uses those delays to reconstruct the original timing.

The handler also accesses the editor state from the Playground context, allowing it to associate each evaluation with the current session state.

We added four icons to the Playground toolbar: start, stop, clear, and open session. The open session button displays the recorded evaluations in a new Playground, where they can be inspected, replayed, or edited into a standalone script.

The design separates recording logic from execution semantics, avoiding invasive changes to the language infrastructure.

4. Conclusion

Using Method Proxies to implement a recorder for live coding performances in Coypu required only a small number of classes and methods, and was achieved without modifying the Coypu codebase. The recorder captures every expression evaluated in the Playground, which opens up possibilities beyond music. Any live coding practice conducted within a Pharo Playground can be recorded with the same mechanism, including, for example, graphics rendered with Bloc. The tool can equally serve as a general prototyping recorder, capturing the sequence of evaluations made during an exploratory session in Pharo. The library ships with a test suite and integrates without issues into the existing environment. Future work will focus on exposing a simpler API within the Phizura package, allowing users to extend the instrumentation layer. This will enable the definition of custom actions triggered either by evaluated expressions or by specific methods within a user's own library, giving live coders finer control over what is captured and how it is processed.

References

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